N100 Financial Platform

Analyst Guide — Sprint 6 Release

Covers ETL, the ratio & screener engines, KMeans clustering, the REST API, and how to interpret every generated report. Database snapshot: 92 companies, 1065 financial-ratio records.

1. Platform Overview

The N100 Financial Platform ingests annual-report-derived fundamentals for the Nifty 100 universe, computes a standard set of financial ratios, and layers screening, peer, cash-flow intelligence, clustering, and reporting modules on top of a single SQLite database.

Current database volume:

Table	Row Count
companies	92
sectors	92
profitandloss	1276
balancesheet	1312
cashflow	1187
financial_ratios	1065
market_cap	552
peer_groups	56
documents	1585

2. ETL & Data Quality

Source Excel workbooks are loaded, cleaned and normalised into the SQLite schema by the modules in src/etl/. Every load pass runs 16 data-quality rules (DQ-01 through DQ-16), covering primary-key uniqueness, referential integrity to the companies table, balance-sheet consistency (assets ≈ liabilities), sign checks on sales/expenses, and completeness checks on year, website, close price, sector and market cap fields.

Rule severities:

• **CRITICAL** — duplicate keys, unknown company_id, duplicate company name. These block a clean load.
• **WARNING** — value-level issues (missing fields, sign violations, balance-sheet drift) that are logged for manual review but do not block the load.

3. Ratio Engine & Screener

src/analytics/ratio_engine.py computes profitability, leverage, efficiency and cash-flow ratios for every company-year. The screener (src/screener/engine.py) applies threshold-based strategies defined in src/config/screener_config.yaml. The six built-in strategies are:

Strategy	Key Thresholds
quality_compounder	ROE ≥ 15%, D/E ≤ 1, FCF ≥ 0, Revenue CAGR ≥ 10%
value_pick	P/E ≤ 15, P/B ≤ 2, ROE ≥ 15%, D/E ≤ 1, Div. Yield ≥ 1%
growth_accelerator	PAT CAGR ≥ 20%, Revenue CAGR ≥ 15%, D/E ≤ 2
dividend_champion	Div. Yield ≥ 2.5%, Payout ≤ 70%, ROE ≥ 10%, FCF ≥ 0
debt_free_bluechip	D/E = 0, ROE ≥ 12%, Sales ≥ ■5,000 Cr
turnaround_watch	Revenue CAGR ≥ 10%, FCF ≥ 0, D/E ≤ 2

4. KMeans Clustering (Sprint 6)

src/analytics/clustering.py groups every company into 5 quantitative clusters using return_on_equity_pct, debt_to_equity, revenue_cagr_5yr, fcf_cagr_5yr and operating_profit_margin_pct. Missing values are imputed with the company's sector median (falling back to the portfolio-wide median). Features are standardised before fitting KMeans(n_clusters=5, random_state=42).

src/analytics/cluster_profiling.py then profiles each cluster's mean/median feature values and ranks them on a composite quality score (ROE + margin – leverage, all z-scored) and a growth score (revenue + FCF CAGR, z-scored) to assign one of five readable names:

• High-Quality Compounders — best quality score, above-median growth
• Distressed or Turnaround — worst quality score, below-median growth
• Emerging Growth — highest growth score among the remainder
• Defensive Dividend Payers — lowest leverage among the remainder
• Value Cyclicals — everything left over

Outputs: reports/elbow_plot.png (k=2..10 inertia curve), output/cluster_labels.csv, reports/correlation_heatmap.png, output/outlier_report.csv (|z-score| > 3), and output/portfolio_stats.csv (P10-P90, mean, std per feature).

5. REST API

The FastAPI layer in src/api/ exposes every analytics module under the /api/v1 prefix, with CORS and request-logging middleware enabled. Interactive documentation is available at /docs (Swagger UI) once the service is running:

```
uvicorn src.api.main:app --reload
```

Router	Base Path	Purpose
health	/api/v1/health	Status, version, uptime, DB row counts
companies	/api/v1/companies	Profile, P&L, BS, cash flow, ratios, tearsheet
screener	/api/v1/screener	Named strategies + ad-hoc filters
sectors	/api/v1/sectors	Sector roll-ups and members
peers	/api/v1/peers	Peer groups and percentile rankings
valuation	/api/v1/valuation	Market cap / PE / PB history and rankings
portfolio	/api/v1/portfolio	Cluster composition, stats, outliers
documents	/api/v1/documents	Annual report links per company

6. Reading the Generated Reports

• **Company Tearsheets** (reports/tearsheets/*.pdf) — 2-page single-company summary: KPI tiles, capital-allocation pattern badge, revenue/profit charts, ROE/ROCE trend, balance sheet and cash-flow charts, pros & cons.
• **Sector Reports** — sector-level aggregation of the same KPI set, used to benchmark a company against its sector.
• **Portfolio Report** — cross-sector summary for a full holdings list.
• **Elbow Plot** — validates that 5 is a reasonable cluster count; inertia should visibly bend around k=5.
• **Correlation Heatmap** — checks for multicollinearity among the 5 clustering features before trusting cluster separations.
• **Outlier Report** — companies whose reported ratios are statistically extreme ($|z| > 3$) — always sanity-check these against the source annual report before acting on them.

7. Operational Checklist

Before relying on any report for a decision:

1. Confirm the ETL run completed with 0 CRITICAL DQ failures.
2. Re-run src/analytics/clustering.py and cluster_profiling.py after any data refresh — cluster assignments are not persisted across data versions automatically.
3. Check output/outlier_report.csv for the companies you are analysing before trusting their ratios at face value.
4. Confirm docs/openapi.json is regenerated (python -m src.api.export_api_docs) after any router change.
5. Run the full test suite (pytest) — 0 failures required before shipping.

8. Ratio Glossary

Definitions for every ratio referenced in this guide, as computed by src/analytics/ratio_engine.py:

Ratio	Definition
ROE (%)	Net Profit ÷ Shareholder Equity × 100
Debt-to-Equity	Total Debt ÷ Shareholder Equity
Operating Profit Margin (%)	Operating Profit ÷ Sales × 100
Interest Coverage	Operating Profit ÷ Interest Expense
Asset Turnover	Sales ÷ Total Assets
Free Cash Flow (■ Cr)	Cash from Operations – Capex
Revenue / FCF / PAT CAGR (5yr)	Compound annual growth rate between the earliest and latest available reporting period
Dividend Payout Ratio (%)	Dividends Paid ÷ Net Profit × 100
Composite Quality Score	Weighted blend of profitability, leverage and growth ratios used for quick ranking

9. Appendix — Current Cluster Profile

Snapshot of the live cluster assignment as of the most recent run of the Day 36/37 pipeline (output/cluster_labels.csv):

Cluster Name	Company Count
Value Cyclicals	55
High-Quality Compounders	15
Emerging Growth	13
Distressed or Turnaround	7
Defensive Dividend Payers	2

For per-company cluster assignment and distance-from-centroid, query GET /api/v1/portfolio/clusters or open output/cluster_labels.csv directly.